Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER ARCHITECTURE

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) Assume suitable data wherever necessary.

(2) All questions are compulsory.

(3) Neat sketches and block diagrams are expected wherever necessary.

1. (a) Implement the given functions using PAL and PLA :

$f_1(a, b, c, d) = \pi M(1, 2, 3, 8, 12, 14, 15) \cdot d(0, 7, 10, 13)$

$f_2(a, b, c, d) = \Sigma m(1, 5, 7, 8, 9, 10, 11, 12, 14)$. 10(CO4)

2. (a) Perform Booth Multiplication algorithm on $(-25)_{10}$ and $(41)_{10}$. 5(CO2)

(b) Convert the given numbers into IEEE 754 single precision floating point binary representation :

(i) $(-210 \cdot 25)_8$

(ii) $(1C23 \cdot F1)_{16}$ 5(CO3)

3. Design Control unit for Full adder in a Computer system using Delay Element Method and State Table Method. 10(CO5)

4. (a) Describe Priority Control Function in a Bus Control Unit. 5(CO1)

(b) Perform division on $(50)_{10}$ by $(3)_{10}$ using restoring algorithm. 5(CO2)

5. (a) Consider a paging system in which main memory has the capacity of three pages. The execution of a program Q requires reference to five distinct pages Pi where i = 1, 2, 3, 4, 5 and i is the page address. The page stream formed by executing Q is :

2 3 1 1 5 1 4 4 3 2 5 2

Assess how pages are assigned to main memory using FIFO, LRU and OPT replacement policies. 5(CO3)

(b) Discuss Flynn's classification of parallelism in a computer system. 5(CO1)

6. (a) Analyze internal organization of CPU with cache memory. 5(CO1)

(b) Discuss locality of reference with respect to virtual memories along with diagram. 5(CO1)